

30 yrs

building hard tech

6 cos

founded / co-founded

400 +

Falcon launches

7M +

Tesla vehicles

Summary

Founder and engineer across SpaceX, Tesla, xAI, Neuralink, The Boring Company, and X. 30 years running companies where the product is hard and the supply chain is physical. Long arc: move civilization to multi-planetary, sustainable energy, beneficial AI. Core depth in rocket engineering, mass manufacturing, capital markets, and first-principles problem decomposition.

Experience

1995 - Present · six companies, one reusable rocket, one mass-market EV

1995 Internet era

Zip2 -> X.com -> PayPal. Sold two internet companies before age 32.

2002 Space + autos

SpaceX, Tesla, SolarCity. Built physical infrastructure others called impossible.

2023 AI era

xAI, Neuralink scaling, X platform ownership. Bet on AGI and BCI.

SpaceX · Space launch · reusable rocketsFOUNDER

ROLE

Founded 2002. CEO and Chief Engineer from day one. First private company to orbit (2008), dock ISS (2012), land a booster (2015), and fly crew (2020).

ACTIONS

Full vertical integration from propellant to avionics. Merlin and Raptor engines in-house. Reusable-booster thesis pursued through 12 failed landings. Launch cadence: 1/year (2008) to every 2-3 days (2024).

IMPACT

Falcon 9: 400+ launches, 95%+ reuse, 10x cost reduction. Starlink: 7,000+ satellites, largest constellation ever. Valued at \$350B (2024).

Tesla · EVs · energy · autonomyCEO

ROLE

Lead investor 2004, CEO from 2008 during near-bankruptcy. Built from a two-car project into the world's most valuable automaker.

ACTIONS

Master Plan framework: each model funds the next. Four Gigafactories on three continents. FSD shipped as continuous software, not fixed release. Vertically integrated batteries, motors, silicon, charging.

IMPACT

7M+ vehicles delivered (1.8M in 2023), first EV to cross 20% gross margin, first automaker past \$1T. Every major OEM announced EV transition within 5 years of Model 3.

xAI · Frontier AI · foundation modelsFOUNDER

ROLE

Founded 2023 to understand the true nature of the universe. Shipped Grok within 6 months, one of the fastest foundation-model companies to production.

ACTIONS

Built Colossus (100K H100 GPUs) in 122 days from concept to training start. Integrated Grok directly into X for real-time data access and instant distribution to 500M+ users.

IMPACT

Colossus in 122 days, a pace the industry called implausible. Grok reached ChatGPT-level benchmarks within 18 months. xAI valued at \$50B (2024).

Neuralink · Brain-computer interfaces · BCI implantsFOUNDER

ROLE

Founded 2016 to build high-bandwidth brain-computer interfaces. Co-leads engineering on the N1 implant, surgical robot, and electrode array design.

ACTIONS

Shipped N1 chip with 1,024 electrodes and precision surgical robot. FDA Breakthrough Device designation and IDE approval (2023) after five years of primate trials.

IMPACT

First human implanted January 2024; full cursor control in 30 days. Second patient mid-2024, expanding to speech restoration. Valued at \$5B (2023). Long-term goal: high-bandwidth symbiosis between human cognition and AI.

First-principles founder across hard tech. Approach: start from the physics, question every received cost assumption, prototype in public. Public reach: 214M followers on X · 125 followers on GitHub.

SpaceXLaunch · Starlink · Starship* \$350B

TeslaEVs · energy · FSD* \$900B

CADENCE The Falcon 9 booster that landed on 22 December 2015 was the first orbital-class rocket to return intact and fly again. The industry assumption that rockets were expendable was gone within 24 months.

Judgment & Conviction

1999 Bet on online banking	2008 CEO during near-bankruptcy	2015 Committed to reusable boosters
Founded X.com while consumer online banking was niche; merger with Confinity became PayPal. Sold to eBay for \$1.5B, with proceeds recycled into SpaceX and Tesla within two years.	Took Tesla CEO during the 2008 crisis, three weeks from insolvency. Closed a Mercedes partnership that funded Model S, enabling every subsequent Tesla product.	Pushed Falcon 9 reusability after 12 consecutive landing failures when board and industry said stop. Succeeded on attempt 13; reusability is now the industry baseline.

Public Impact

X · @elonmusk 214M followers Direct channel to 214M followers. Each engineering post, launch video, or product hint moves markets within hours.

Selected writing	Invited talks
Master Plan Part Deux2016.07 Tesla blog, 40M+ reads; reframed EV strategy across the industry	Making Humans Multi-Planetary2016.09 IAC Guadalajara · 2016
First Principles Thinking in Engineering2013.03 TED Talk · 5.6M views on youtube.com	Starship and Mars2024.04 Starbase Texas · 2024

Core Skills

ROCKET · ENGINE DESIGN	Led Merlin, Raptor, Starship architecture; first reusable orbital booster shipped by any org.
MANUFACTURING · SCALE	Built Tesla Fremont, Shanghai, Berlin, Austin; SpaceX assembly scaled from 0 to 100+ launches/year .
CAPITAL · CRISIS MGMT	Closed Daimler investment when Tesla was weeks from insolvency ; survived five near-bankruptcies across companies.
PUBLIC COMMUNICATION	214M followers on X; announcements routinely move the S&P 500 component price of his companies.
FIRST PRINCIPLES · COST	Questioning every accepted cost assumption (rockets, batteries, tunneling) has driven 10× cost reductions across four industries.

Education

University of Pennsylvania · Wharton, B.S. Economics · B.A. Physics

1992 - 1997